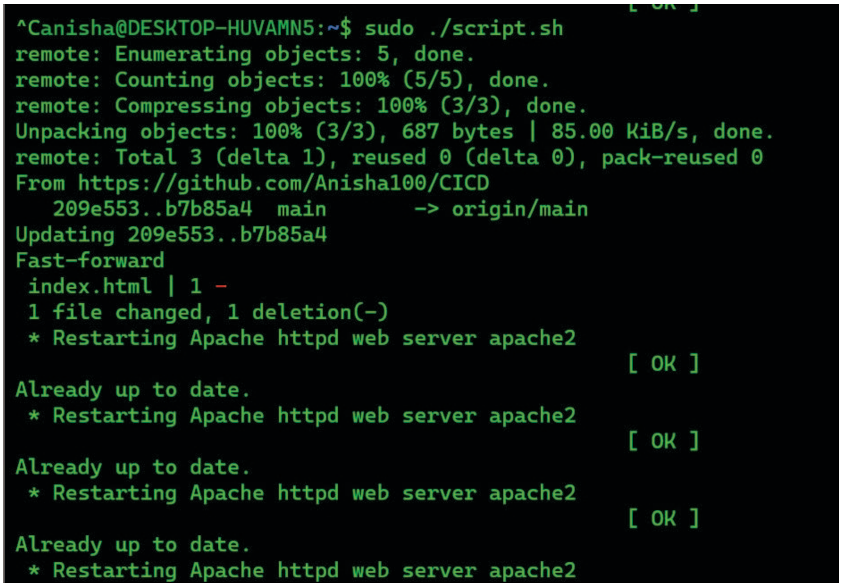
Creating a simple CI/CD bash script

Let us take a simple HTML file deployed on Apache server for demonstration, where we create a folder named *CICD* which has been cloned from GitHub, and add two files in it, one of which is a JavaScript file called *sample.js*. This file contains a timeout for 3000 seconds, and reloads the web page using the *reload ()* function so that the user is not required to refresh the web page to notice any new changes. Next, we create the *index.html* file, which contains the information required to be displayed on the web page.

```
anisha@DESKTOP-HUVAMN5:~/CICD$ cat
sample.js
setTimeout(function(){
window.location.reload();
},3000);
anisha@DESKTOP-HUVAMN5:~/CICD$ cat
index.html
<!DOCTYPE html>
<html>
<script src=”sample.js”></script>
<body>
<h1> This is a sample html page</h1>
<p>This is deployed on Apache server</
p>
<marquee> Test </marquee>
</body>
</html
anisha@DESKTOP-HUVAMN5:~/CICD$ ls
index.html sample.js
anisha@DESKTOP-HUVAMN5:~/CICD$
```

Let us now create a shell script containing the CI/CD bash script you want to be executed for your deployment. The code given below shows that I am running an infinite loop, with a sleep interval of 3 seconds. In case there are multiple developers working on the same project, this file will be constantly running every 3 seconds, making sure that the code from GitHub is uploaded using the command *git pull*.

The next line copies this bash file to the */var/www/html* folder, where all the Apache HTML files are to be stored.



```
^Canisha@DESKTOP-HUVAMN5:~$ sudo ./script.sh
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
Unpacking objects: 100% (3/3), 687 bytes | 85.00 KiB/s, done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
From https://github.com/Anisha100/CICD
   209e553..b7b85a4  main       -> origin/main
Updating 209e553..b7b85a4
Fast-forward
 index.html | 1 -
 1 file changed, 1 deletion(-)
 * Restarting Apache httpd web server apache2
                                                    [ OK ]
Already up to date.
 * Restarting Apache httpd web server apache2
                                                    [ OK ]
Already up to date.
 * Restarting Apache httpd web server apache2
                                                    [ OK ]
Already up to date.
 * Restarting Apache httpd web server apache2
```

Figure 1: Running *script.sh*

Figure 2: Sample HTML page

Then the line removes the error logs for the CICD repository and restarts the Apache server. This goes on till the developer manually stops the infinite loop. After this, we run the shell script, which shows the changes made, and performs all the instructions specified in the script. We can see the server restarting after every 3 seconds and automatically updating the new changes without the need for human intervention.

```
anisha@DESKTOP-HUVAMN5:~$ cat script.sh
cd CICD

while true
do
git pull
sudo cp
/var/www/html
sudo rm -f /var/logs/apache2/error.log
sudo service apache2 restart
sleep 3
done
```

Figure 1 shows the output when the above script is executed.

You can clone your repository or work on your local system, download a server like Apache or Nginx and start it to see the HTML page running on the server when you type *http://localhost.*

This runs on any VM using a Linux based distro, such as using cloud shells for instances and VMs. Figure 2 shows the web page running.

I hope this short tutorial will make things easier for developers by automating the mundane and tedious tasks of infrastructure management. END

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The author is a cyber security researcher and an active contributor to open source communities and repositories. She is interested in developing novel, scalable and secure systems.